Environmental Product Declaration

In accordance with EN 15804:2012 + A2:2019/AC:2021, ISO 21930:2017 and ISO 14025 for:

IT - Robilante - Silica VVR WS (Silica Sand Wet)

EPD Owner
Sibelco Group
Reference Facility: Robilante (Italy)

Programme:	The International EPD® System. www.environdec.com
Programme operator:	EPD International AB
EPD registration number:	EPD-IES-0028701:002
Version date:	2026-02-24
Validity date:	2031-02-11
Revision date:	NA

An EPD may be updated or republished if conditions change. To find the latest version of the EPD and to confirm its validity, see www.environdec.com.

Programme information

EPD process certification involves an accredited certification body certifying and periodically auditing the EPD process and conducting external and independent verification of EPDs that are regularly published. More information can be found in the General Programme Instructions on www.envrondec.com

Programme used: The International EPD® System. For more information see www.environdec.com

Address of programme operator: EPD International AB, Box 210 60, SE-100 31 Stockholm, Sweden, E-mail:info@environdec.com

Product / product family name and manufacturer represented: Silica VVR WS (Silica Sand Wet) product, manufactured by Sibelco in the Robilante facility.

Declaration issued: 2026-02-12

Valid until: 2031-02-11

Owner of the declaration: SCR-Sibelco NV - Plantin en Moretuslei 1A, 2018 Antwerpen, Belgium
Tel. +32 33931010 E- mail. info@sibelco.com
The EPD owner has the sole ownership, liability and responsibility for the EPD.

CEN standard EN 15804 served as the core Product Category Rules–PCR	
PCR:	PCR 2019:14 Construction products, Version 2.0.1
Product group classification:	The UN CPC code of the product is 153
PCR review was conducted by:	The Technical Committee of the International EPD® System. Chair: Rob Rouwette, Co-Chair: Noa Meron. email: support@environdec.com
Independent verification of the declaration and data, according to ISO 14025:	☐ Individual EPD verification without a pre-verified LCA/EPD tool ☐ Individual EPD verification with a pre-verified LCA/EPD tool ☒ EPD process certification without a LCA/EPD pre-verified toll ☐ EPD process certification with a pre-verified LCA/EPD tool ☐ Fully pre-verified EPD tool
Third party verifier:	SGS ICS Italia S.r.l Via Caldera 21, 20153 Milano (Italy)
Accredited by:	Accredia - n° 02242

According to EN 15804, EPDs of construction products may not be comparable if they do not comply with this standard.

EPDs within the same product category but registered in different EPD programmes may not be comparable. For two EPDs to be comparable, they shall be based on the same PCR (including the same version number up to the first two digits) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterisation factors); have equivalent content declarations; and be valid at the time of comparison. For further information about comparability, see EN 15804 and ISO 14025.

Underlying LCA study carried out by: Sibelco Group

EPD prepared by: Sibelco Group

About the Company

General

Sibelco is a global material solutions company that sources, transforms and distributes specialty industrial minerals – particularly silica, clays, felspathic and olivine – and in glass recycling. Sibelco delivers solutions for society and supports the progress of modern life, from drinking water sanitation to solar technology and from flame retardants to smartphone screens. Industries served are as diverse as glass, ceramics, construction, coatings, polymers and water purification. Founded back in 1872, Sibelco has grown into a multinational business with operations in around 30 countries and an extensive multi-mineral portfolio. The goal is to deliver long-term social, environmental and economic value for stakeholders, continuously guided by the purpose: Material solutions advancing life. Refer to Sibelco website for additional info.

Reference Facility

Robilante site located in the North-Western part of Italy produces silica, mainly for the construction, glass, sports & leisure and agriculture industries.

Product description

Reference Product

Silica (SiO_2) is the term used to describe a group of minerals composed of oxygen and silicon. Whilst it is the world's second most abundant mineral, products containing at least 98% SiO_2 are needed for industrial use. As a hard, chemically inert mineral with a high melting point, silica delivers a range of benefits across multiple applications: glass, ceramics, construction, metallurgy, sports turf and many others.
For more information refer to the Materials section of Sibelco website.

Main **properties** of the reference product are reported in the table below. For additional details, refer to the Materials section of Sibelco website and to the Sibelco Shop Online Website.

Table 1. - Properties of Silica VVR WS (Silica Sand Wet)

Property	Value	Method
Moisture content	20%	Microwave oven and centesimal scale
Particle Size distribution (D50)	9.0 um	Sieving

Main **applications** of the reference product are indicated in the table below. For additional details, refer to the Materials section of Sibelco website and to the Sibelco Shop Online Website.

Table 2. - Application of Silica VVR WS (Silica Sand Wet)

Application	Description
Construction	To enhance the performance of a range of building materials, from faster-setting cement and more flame-resistant fiber cement boards.
Tiles & Engobes	To achieve improved technical solutions such as increased mechanical strength and lower weight, whilst at the same time managing manufacturing costs.
Sport & Leisure	To ensure that the surface is always in peak condition with our wide range of high-performance sands, top dressing and rootzone materials, engineered to your specifications.

Main **shipping size and methods** are listed in the table below. For additional details, refer to the Sibelco Shop Online Website

Table 3. - Shipping size and method of Silica VVR WS (Silica Sand Wet)

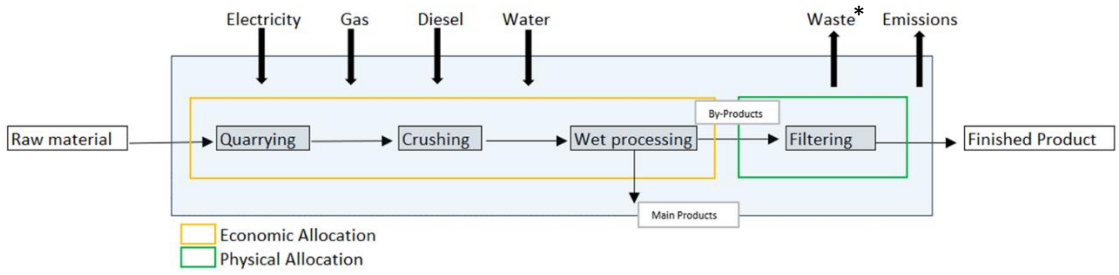
Method	Size (kg)
Bulk	30.000 kg

The product is distributed worldwide so the geographical scope is global.

Process

Raw material is extracted from the quarry, by explosive and mobile equipment. Rocks are transported to the processing plant and reduced in size via crushing and milling steps, before entering the washing plant where the sand is washed, screened and cleaned. Main products and by-products are extracted at the same time. Thus, the materials are filtered reducing water content and stocked in silos. They are shipped in bulk.
Part of the electricity is provided by a regenerative gravity generator, a solar panels installed on the ground and a gas-fired CHP.

Figure 1 - Flow chart of Silica VVR WS (Silica Sand Wet) production



* EoW state according to Annex 4 of PCR 2019:14 Construction products, Version 2.0.1 at the gate of waste disposer for the waste recycling stream.

Scope

The LCA is based on 2025 production data for Robilante manufacturing site in Italy, for Silica VVR WS (Silica Sand Wet). This EPD covers the life cycle stages of product stage and construction stage for Silica VVR WS (Silica Sand Wet) sold and used in different markets and countries. Use stage, EoL stage and resources recovery stage were not considered in this study.
All processes were modelled using the Ecoinvent database v 3.11; SimaPro 10.2.0.0 and a customized tool for calculations.

Geographical scope: Italy (A1, A2, A3), Global (A5).

All data refer to 2025, except for air emissions, water, waste generated which refer to 2024. The flows of emissions, waste, water are constant over time and the 2024 data derive from surveys which are collected from the sites and validated centrally during H2 of the 2025.

System boundary

The selected system boundary type is e) of Section 2.2.2 of the PCR: from cradle to gate with options (A1–A3 and additional modules). The additional module considered is A5. The system boundary covers the Product stage (raw material supply as A1; transport up to the factory gate as A2; manufacturing as A3), as well as the Construction process stage (A5). The remaining modules were not declared since:

- the product or material is physically integrated with other products during installation so they cannot be physically separated from them at end of life;
- the product or material is no longer identifiable at end of life as a result of a physical or chemical transformation process;
- the product or material does not contain biogenic carbon.
- the EPD is not intended to be used for business-to-consumer communication.

The system boundaries in tabular form for all modules are shown in the Table 4 below.

Table 4 - System boundary chosen for the LCA (X-module included in LCA. ND – module not included).

	PRODUCT STAGE			CONSTRUCTION STAGE		USE STAGE							END OF LIFE STAGE				RESOURCE RECOVERY STAGE
	Raw material supply	Transport	Manufacturing	Transport from the gate to the site	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse- Recovery- Recycling- potential
MODULE	**A1**	**A2**	**A3**	A4	**A5**	B1	B2	B3	B4	B5	B6	B7	**C1**	**C2**	**C3**	**C4**	**D**
Module declared	X	X	X	ND	X	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND
Geography	IT	IT	IT	-	GLO	-	-	-	-	-	-	-	-	-	-	-	-
Specific data	>90%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – products	0%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – sites	0%			-	-	-	-	-	-	-	-	-	-	-	-	-	-

Product stage, A1-A3

The product stage is subdivided into three modules; A1, A2 and A3 respectively “raw material supply”, “transport” and “manufacturing”.

A1, raw material supply
This includes the extraction and processing of all raw materials and energy which occur upstream from the manufacturing process.

A2, transport to the manufacturer
Raw materials and additives are supplied by truck from manufacturers within Italy or from neighbouring countries.

A3, manufacturing
This module includes the manufacture of products and the production of packaging material. The processing of any waste arising from this stage is also included.

Product stage, A5

A5, Construction installation
Thie module includes packaging End-of-Life, for balancing out biogenic carbon.

Functional unit/declared unit

The declared unit (DU) is 1000 kg of Silica VVR WS (Silica Sand Wet) for product application, as shown in table 2.

Product composition

The average composition of 1000 kg of product is reported in Table 5.

Table 5 - Content declaration of product

Product components	Product weight [%]	Product weight [kg]	Post-consumer [%]	Pre-consumer [%]	Biogenic, weight [%] *
Silica	100%	1000	0%	0%	0%
Total	100%	1000	0%	0%	0%

*the biogenic content is set to zero according to the PCR requirement.

Declaration of data sources, reference years, data categories, and share of primary data, see the table 6 below

Table 6 - Declaration of data sources, reference years, data categories, and share of primary data

Process	Source type	Source	Reference year	Data category	Share of primary data, of GWP-GHG results for A1-A3
Generation of electricity used in manufacturing of product	Database	Ecoinvent v3.11	2024	Secondary data	77,7%
Raw material extraction	Collected data	EPD owner	2024	Primary data	4,3%
Raw material supply	Collected data	Ecoinvent v3.11	2024	Primary data, secondary data	0,0%
Packaging raw material	Database	Ecoinvent v3.11	2024	Primary data, secondary data	0,0%
Mobile equipment internal and external	Collected data, Database	Ecoinvent v3.11	2024	Primary data, secondary data	0,0%
Heat production	Collected data	EPD owner	2024	Primary data	0,0%
Emission to air	Collected data	EPD owner	2024	Primary data	0,0%
Other processes	Databases	Ecoinvent v3.11	2024	Secondary data	0,0%
Total share of primary data, of GWP-GHG results for A1-A3					**82,1%**

The share of primary data is calculated based on GWP-GHG results. It is a simplified indicator for data quality that supports the use of more primary data, to increase the representativeness of and comparability between EPDs. Note that the indicator does not capture all relevant aspects of data quality and is not comparable across product categories.

Primary data specific for the actual production have been collected from Sibelco, referred to year 2025. Secondary data have been obtained from Ecoinvent 3.11 taking into consideration geographical, technical, and temporal representativeness of the data. Quality level concerning datasets used in the EPD can be considered as “very good” according to Annex E of the EN 15804.

Additional information

The reference product Silica VVR WS (Silica Sand Wet) is **100% a by-product**.
The economic allocation adopted for the first process stages reflects the economic value of the “Fines” produced after the Washing stage in combination with the other main products for which the production process is intent for. Then, the physical allocation is adopted to properly allocated the impacts of the following steps entirely dedicated to the reference product.

Further information

Further information can be found through the enquiry desk:
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Environmental performance

In the following tables, the environmental impacts per declared unit and per life cycle stage are reported for environmental impact categories recommended by the International EPD® System (https://www.environdec.com/resources/indicators) and those indicated in in PCR 2019:14 version 2.0.1 for Construction Products and construction services.

Impact categories are the one required by Version 2.0 of the EPD default list, with characterisation factors from "EN 15804 reference package" provided by JRC and based on version 3.1 of the reference package for CFs used in the PEF framework (EF 3.1).

For the impact categories Resource use, minerals and metals and Resource use, fossils, the following disclaimer applies: the results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is a limited experienced with the indicator.

For the impact categories in Table 6 the estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

Table 7 - LCA results of potential environmental impact – core indicators referred to declared unit

Silica VVR WS (Silica Sand Wet)		PRODUCT LIFE CYCLE INFORMATION	
Reference PCR		PCR 2019:14 Construction products, Version 2.0.1	
SIZE (kg)	1000 kg	PRODUCT STAGE	CONSTRUCTION PROCESS STAGE
		A1-A3	A5
		Total	Construction-installation process
CORE ENVIRONMENTAL IMPACT INDICATORS			
INDICATOR	UNIT		
Climate change (additional - NO CO2 Bio)	kg CO_2 eq	1,423E+01	0,000E+00
Climate change (total)	kg CO_2 eq	1,423E+01	0,000E+00
Climate change - Fossil	kg CO_2 eq	1,421E+01	0,000E+00
Climate change - Biogenic	kg CO_2 eq	2,217E-02	0,000E+00
Climate change - Land use and LU change	kg CO_2 eq	3,118E-03	0,000E+00
Ozone depletion	kg CFC11 eq	9,695E-09	0,000E+00
Acidification	mol H+ eq	4,402E-02	0,000E+00
Eutrophication, freshwater	kg P eq	1,824E-03	0,000E+00
Eutrophication, marine	kg N eq	1,552E-02	0,000E+00
Eutrophication, terrestrial	mol N eq	1,690E-01	0,000E+00
Photochemical ozone formation	kg NMVOC eq	6,716E-02	0,000E+00
Resource use, minerals and metals	kg Sb eq	5,704E-05	0,000E+00
Resource use, fossils	MJ	2,292E+01	0,000E+00
Water use	m^3 depriv.	6,254E+00	0,000E+00

Table 8 - LCA results of additional indicators referred to the declared unit

Silica VVR WS (Silica Sand Wet)		PRODUCT LIFE CYCLE INFORMATION	
Reference PCR		PCR 2019:14 Construction products, Version 2.0.1	
SIZE (kg)	1000 kg	PRODUCT STAGE	CONSTRUCTION PROCESS STAGE
		A1-A3	A5
		Total	Construction-installation process
ADDITIONAL ENVIRONMENTAL IMPACT INDICATORS			
INDICATOR	UNIT		
Particulate matter	disease inc.	5,824E-07	0,000E+00
Ionising radiation	kBq U-235 eq	2,445E-01	0,000E+00
Ecotoxicity, freshwater	CTUe	1,996E-03	0,000E+00
Human toxicity, cancer	CTUh	3,061E-09	0,000E+00
Human toxicity, non-cancer	CTUh	9,331E-08	0,000E+00
Land use	Pt	2,452E+02	0,000E+00

Table 9 - LCA results of use of resources referred to the declared unit

Silica VVR WS (Silica Sand Wet)		PRODUCT LIFE CYCLE INFORMATION	
Reference PCR		PCR 2019:14 Construction products, Version 2.0.1	
SIZE (kg)	1000 kg	PRODUCT STAGE	CONSTRUCTION PROCESS STAGE
		A1-A3	A5
		Total	Construction-installation process
PARAMETERS DESCRIBING RESOURCE USE			
INDICATOR[1]	UNIT		
Primary en res, Ren, Energy carrier	MJ	5,158E+00	0,000E+00
Primary en res, Ren, Raw material	MJ	0,000E+00	0,000E+00
Primary en res, Ren, Total	MJ	5,158E+00	0,000E+00
Primary en res, Non-ren, Energy carrier	MJ	2,292E+01	0,000E+00
Primary en res, Non-ren, Raw material	MJ	0,000E+00	0,000E+00
Primary en res, Non-ren, Total	MJ	2,292E+01	0,000E+00
Use of secondary material	kg	0,000E+00	0,000E+00
Use of renewable secondary fuels	MJ	0,000E+00	0,000E+00
Use of non-renewable secondary fuels	MJ	0,000E+00	0,000E+00
Net use of fresh water	m3	4,472E+00	0,000E+00

[1] Option A is applied for energy balancing, and all inherent energy in the product and packaging is assumed to be lost within the system boundary as a conservative scenario.

Table 10 - LCA results of waste categories referred to the declared unit

Silica VVR WS (Silica Sand Wet)		PRODUCT LIFE CYCLE INFORMATION	
Reference PCR		PCR 2019:14 Construction products, Version 2.0.1	
SIZE (kg)	1000 kg	PRODUCT STAGE	CONSTRUCTION PROCESS STAGE
		A1-A3	A5
		Total	Construction-installation process
PARAMETERS DESCRIBING WASTE CATEGORIES			
INDICATOR	UNIT		
Non hazardous waste	kg	7,226E-01	0,000E+00
Hazardous waste	kg	1,128E-03	0,000E+00
Radioactive waste	kg	6,132E-05	0,000E+00

Table 11 - LCA results of output flows referred to the declared unit

Silica VVR WS (Silica Sand Wet)		PRODUCT LIFE CYCLE INFORMATION	
Reference PCR		PCR 2019:14 Construction products, Version 2.0.1	
SIZE (kg)	1000 kg	PRODUCT STAGE	CONSTRUCTION PROCESS STAGE
		A1-A3	A5
		Total	Construction-installation process
PARAMETERS DESCRIBING OUTPUT FLOWS			
INDICATOR	UNIT		
Components for re-use	kg	0,000E+00	0,000E+00
Materials for recycling	kg	7,216E+01	0,000E+00
Materials for energy recovery	kg	0,000E+00	0,000E+00
Exported energy, electricity	MJ	0,000E+00	0,000E+00
Exported energy, thermal	MJ	0,000E+00	0,000E+00

Table 12 - LCA results of biogenic carbon content indicators referred to the declared unit

Silica VVR WS (Silica Sand Wet)		PRODUCT LIFE CYCLE INFORMATION
INFORMATION ON BIOGENIC CARBON CONTENT		
INDICATOR	**UNIT**	
Biogenic carbon content in product	kg C	0E+00
Biogenic carbon content in accompanying packaging	kg C	0,000E+00

Table 13 - LCA results of electricity contribution indicators referred to the declared unit

Silica VVR WS (Silica Sand Wet)		PRODUCT LIFE CYCLE INFORMATION
ELECTRICITY USE IN A1-A3		
INDICATOR	**UNIT**	
Electricity contribution to GWP-GHG in modules A1-A3	%	77,74%
GWP-GHG results for electricity	kg CO2/kWh	0,578

Detailed data of energy sources behind electricity used in manufacturing process and GWP- GHG impact are reported in the following table.

Energy Source	GWP-GHG (kgCOeq/kWh)
Electricity from national grid (Ecoinvent, Italian Residual Mix)	0,680
Solar groud (self-produced)	0,081
CHP	0,597

VERSION HYSTORY

Original version of the EPD, 12/02/2026

“Publication date” change to “Version date”, 24/02/2026

“Valid until” changed to “Validity date”, 24/02/2026

PCR review changed to chair Rob Rouwette (chair) and Noa Meron (co-chair), 24/02/2026

Added “EPD is not intended to be used for business-to-consumer communication” in System boundary, 24/02/2026

Added the Quality level, 24/02/2026

Added the detailed data of energy sources, 24/02/2026

References

EPD International 2021. General Programme Instruction of the International EPD® System, version 5.0.1

GR-QUA-001 EPD process_V2

Ann03-EPD_Structure and functionality of the calculation tool_V1.3.1

Ann04_LCI List_v07

EPD Tool_Sibelco_Robilante_Silica Wet Sand (Silica VVR WS)_20260205_v07

PCR. PCR 2019:14 version 2.0.1. for Construction Products and construction services. IVL Swedish Environmental Research Institute, Swedish Environmental Protection Agency, SP Trä. Swedish Wood Preservation Institute, Swedisol, SCDA. Svenskt Limträ AB. SSAB (2019).

EN 15804:2012+A2:2019/AC:2021 **"**Sustainability of construction works – Environmental product declarations – Core rules for the product category of construction products"

UNI EN ISO 14025:2010. Environmental labels and declarations – Tipe III environmental declarations – Principles and procedures

UNI EN ISO 14040:2021. Environmental management – Life cycle assessment – Principles and framework

UNI EN ISO 14044:2021. Environmental management – Life cycle assessment – Requirements and guidelines

UNI EN ISO 21930:2017. Sustainability in buildings and civil engineering works — Core rules for environmental product declarations of construction products and services

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In the creation of this EPD, Sibelco relied on the technical and methodological support of a research and management consulting team from Tecno s.r.l., , coordinated by **Elena Baldereschi** and **Daniel Borgobello**.

Acronyms

General Acronymus	
EN	European Norm (Standard)
EPD	Environmental Product Declaration
EF	Environmental Footprint
GPI	General Programme Instructions
ISO	International Organization for Standardization
LCA	Life Cycle Assessment
PCR	Product Category Rules
c-PCR	Complementary Product Category Rules
CEN	European Committee for Standardization
CPC	Central product classification
Environmental Impact Indicators (EN 15804)	
GWP	Global warming potential
GWP-luluc	Global warming potential- land use and land use change
ODP	Ozone depletion potential
AP	Acidification Potential
EP	Eutrophication potential (for freshwater, marine, terrestrial compartments)
POCP	Photochemical ozone creation potential
ADP	Abiotic depletion potential
WDP	Water deprivation potential
GWP-GHG	Global warming potential- accounts for all greenhousegases except biogenic CO2 uptake and emissions and biogenic carbon stored in the product and/or the packaging.
PM	Particulate Matter
IRP	Ionizing radiation Potential
ETP-FW	Ecotoxicity Potential- freshwaters
HTP-C	Human Toxicity potential- Cancer
HTP-NC	Human Toxicity potential- Non-Cancer
SPQ	Index of soil quality potential
PERE	Use of primary energy excluding renewable primary energy resources used as raw materials
PERM	Use ofrenewable primary energy resources used as raw materials
PERT	Use of Total use ofrenewable primary energy resources
PENRE	Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials
PENRM	Use of non-renewable primary energy resources used as raw materials
PENRT	Use of Total use of non-renewable primary energy resources
SM	Use of secondary materials
RSF	Use of renewable secondary fuels
NRSF	Use of non-renewable secondary fuels
FW	Use of fresh water (net)
HW	Hazardous Waste (disposed) (kg)
NHW	Non-Hazardous Waste (disposed) (kg)
RW	Radioactive Waste (disposed) (kg)
CFR	Components for Reuse (kg)
MR	Material for Recycling (kg)
MER	Materials for Energy Recovery (kg)

EEE	Exported Energy, Electricity (MJ)
EEE	Exported Energy, Thermal (MJ)

LifeCycle Stages/ Modules

A1	Raw material supply
A2	Transport
A3	Manufacturing
A4	Transport to site
A5	Construction/Installation
B1	Use
B2	Maintenance
B3	Repair
B4	Replacement
B4	Replacement
B5	Refurbishment
B6	Operational energy use
B7	Operational water use
C1	Deconstruction/Demolition
C2	Transport to waste processing
C3	Waste processing
C4	Disposal
D	Reuse-Recovery-Recycling potential Other relevant

Other relevant Terms

SVHC	Substances of Very High Concern
MJ	Megajoule
kg	Kilogram
m^3	Cubic Meter
NMVOC	Non-Methane Volatile Organic Compounds
Sb eq.	Antimony Equivalents
P eq.	Phosphorus Equivalents
N eq.	Nitrogen Equivalents
CFC-11 eq.	Chlorofluorocarbon-11 Equivalents
CO_2 eq.	Carbon Dioxide Equivalents
kg C	Kilograms of Carbon
kg CO_2 eq.	Kilograms of Carbon Dioxide Equivalent
ND	Not Declared